

SMART MARKET

FUNCTIONAL & TECHNICAL REQUIREMENTS DOCUMENT

Prepared by GODWIN TECH SOLUTION

Version 1.0

1. PROJECT OVERVIEW

Smart Market is a multi-vendor e-commerce marketplace built with Next.js and Progressive Web App (PWA) technology.

The platform allows users to create online stores, publish products through paid subscription plans, and complete secure transactions.

The platform integrates an escrow payment system where customer payments are temporarily held by the platform administrator until the buyer and seller confirm successful completion of the transaction.

2. PROJECT OBJECTIVES

- Create a scalable marketplace for multiple sellers.
- Allow merchants to create and manage online stores.
- Generate revenue through subscription-based advertising packages.
- Secure transactions through escrow payment management.
- Provide SEO optimized pages for search engine visibility.
- Integrate an AI chatbot to assist users and improve customer experience.

3. USER TYPES

Visitor:

- Browse products and stores.
- Search products.
- Register an account.

Buyer:

- Manage profile.
- Purchase products.
- Track orders.
- Confirm delivery completion.

Seller / Store Owner:

- Create a store.

- Purchase subscription plans.
- Publish advertisements.
- Manage products and sales.

4. STORE MANAGEMENT

Each seller can create a public store containing:

- Store name.
- Logo and banner.
- Description.
- Location.
- Product categories.
- Contact information.

Each store must have SEO friendly public pages.

5. SUBSCRIPTION AND ADVERTISEMENT SYSTEM

Publishing advertisements requires an active subscription.

Each subscription defines:

- Duration.
- Number of advertisements allowed.
- Available features.

Example:

Starter: 30 days / 10 ads

Business: 90 days / 50 ads

Premium: 365 days / 200 ads

When a subscription expires:

- Existing advertisements remain saved.
- Advertisements lose public visibility.
- Seller can renew existing ads or publish new advertisements after renewing.

6. ADVERTISEMENT MANAGEMENT

Advertisement information:

- Title.
- Description.
- Images.
- Video.
- Price.
- Category.
- Location.
- Product condition.

Advertisement lifecycle:

Draft -> Published -> Active -> Expired -> Renewed -> Archived

7. SEARCH AND SEO REQUIREMENTS

SEO is a core requirement.

The platform must provide:

- SEO friendly URLs.
- Dynamic metadata.
- Product structured data.
- Store structured data.
- Sitemap generation.
- Optimized images.
- Category landing pages.
- Internal linking strategy.

Examples:

smartmarket.com/product/product-name

smartmarket.com/store/store-name

8. MARKETPLACE TRANSACTION FLOW

1. Buyer selects a product.
2. Buyer completes payment.
3. Payment is transferred to Smart Market escrow.
4. Seller delivers product/service.

5. Buyer confirms completion.

6. Funds are released to seller.

9. ESCROW PAYMENT SYSTEM

Smart Market acts as a trusted third party.

Payment flow:

Buyer

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Payment

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Smart Market Escrow

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Completion validation

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Seller payout

The system must support:

- Payment tracking.
- Commission calculation.
- Seller balance management.
- Transaction history.

10. AI CHATBOT

The platform must integrate an AI chatbot capable of:

Customer assistance:

- Answer common questions.
- Guide users through purchasing.
- Explain subscription plans.
- Assist sellers.

Marketplace assistance:

- Product discovery.

- Recommendation assistance.
- Order information support.

Administrative assistance:

- Basic reporting support.
- Customer service assistance.

11. ADMINISTRATION MODULE

Roles:

Super Admin:

- Full platform control.
- Escrow management.
- Subscription management.
- User management.
- Platform configuration.

Accountant:

- Financial reports.
- Payment monitoring.
- Seller payouts.
- Transaction reconciliation.

Customer Service:

- User assistance.
- Complaint management.
- Dispute handling.

12. SELLER DASHBOARD

Dashboard indicators:

- Active advertisements.
- Expired advertisements.
- Subscription status.
- Views.
- Orders.

- Sales revenue.

13. ADMIN DASHBOARD

Key indicators:

- Users.
- Stores.
- Revenue.
- Subscription sales.
- Escrow balance.
- Transactions.
- Disputes.

14. NOTIFICATIONS

Supported notifications:

- Email.
- Push notifications.
- In-platform notifications.

Events:

- Subscription expiration.
- Advertisement expiration.
- New orders.
- Payment confirmation.
- Dispute creation.

15. TECHNICAL REQUIREMENTS

Frontend:

Next.js with Progressive Web App (PWA).

Backend:

Scalable API architecture.

Database:

Relational database management system.

Requirements:

- Secure authentication.
- Role based permissions.
- API integration.
- Cloud storage support.
- Payment gateway integration.

16. MAIN DATABASE ENTITIES

Users

Stores

Subscription Plans

Subscriptions

Advertisements

Categories

Orders

Payments

Escrow Transactions

Wallets

Reviews

Disputes

Notifications

Roles

Permissions

17. DEVELOPMENT PHASES

Phase 1:

- Authentication.
- Stores.
- Subscription system.
- Advertisement management.
- Marketplace.

- Escrow payment.

Phase 2:

- AI chatbot.
- Reviews.
- Advanced analytics.
- Promotions.

Phase 3:

- AI recommendations.
- Advanced marketplace automation.

18. ACCEPTANCE CRITERIA

The project is validated when:

- Users can create accounts.
- Sellers can create stores.
- Sellers can subscribe to plans.
- Advertisement quotas are enforced.
- Expired ads become invisible.
- Buyers can purchase products.
- Escrow payments work correctly.
- Funds are released after confirmation.
- SEO requirements are implemented.
- AI chatbot is operational.

GODWIN TECH SOLUTION

Digital Solutions & Software Engineering